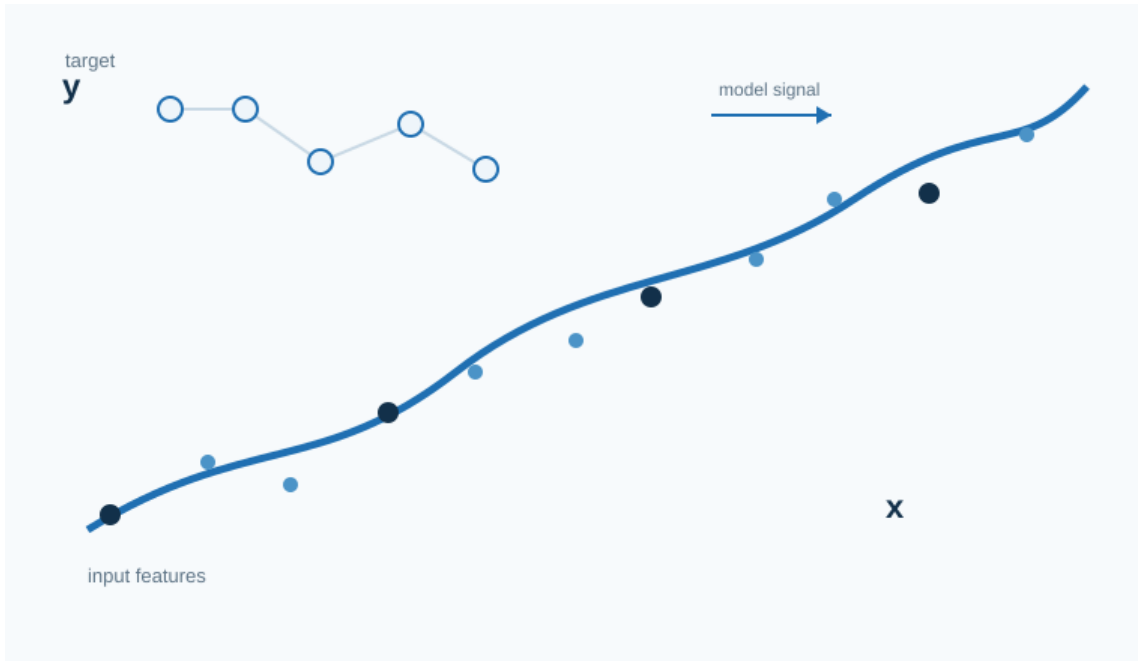


PART A / CONCEPTUAL UNDERSTANDING

SUPERVISED LEARNING

Fundamentals of Regression, Classification, Linear Regression, Model Assumptions and Model Complexity

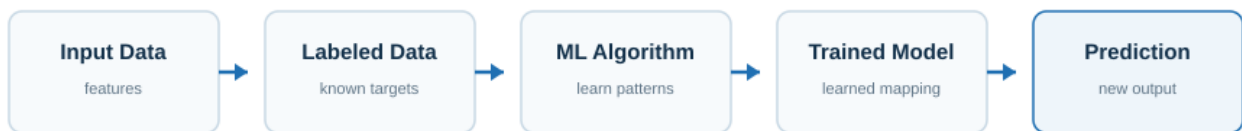


LEARNING NOTE A concise technical reference for supervised learning fundamentals, model assumptions, and model complexity.

01 What are Supervised Learning Algorithms?

Supervised Learning is a type of Machine Learning where a model learns from **labeled data**, meaning the input data already has a known output or target value. The model learns the relationship between input features and the target and then uses it to make predictions on new data. Examples include Linear Regression, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine.

Supervised learning workflow



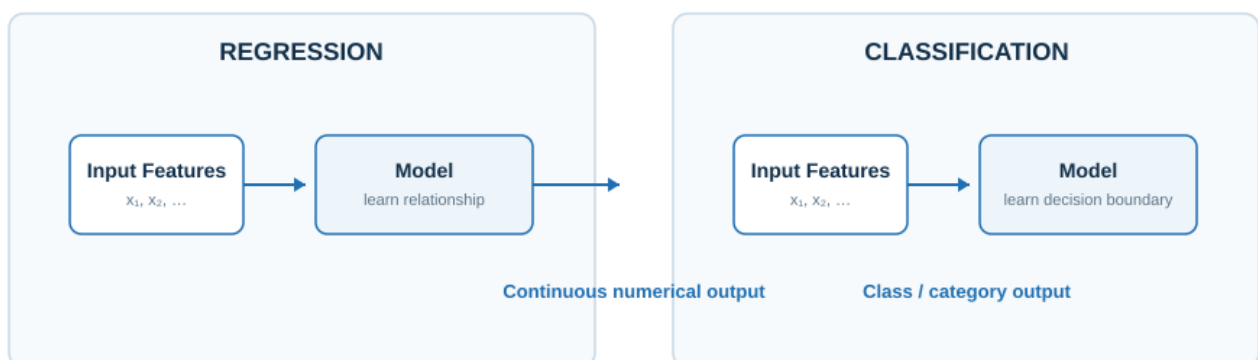
The target value is known during training and used to guide learning.

KEY IDEA The model learns from examples where the correct answer is already known.

Examples: Linear Regression, Logistic Regression, Decision Tree, Random Forest, and Support Vector Machine.

02 Difference between Regression Algorithms vs Classification Algorithms

Regression algorithms are used when the output is a continuous numerical value, such as predicting house price, salary, or temperature. **Classification algorithms** are used when the output belongs to a category or class, such as predicting whether an email is spam or not spam, or whether a customer will default or not default.

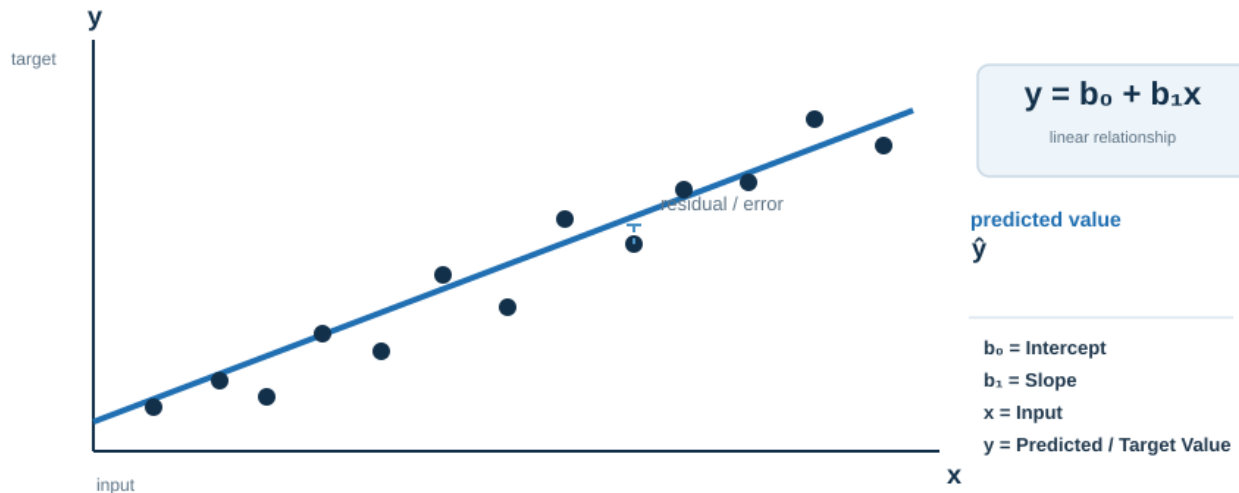


REMEMBER Use regression for “how much?” or “how many?” and classification for “which class?”

03 Explain Simple Linear Regression

Simple Linear Regression is a supervised learning algorithm used to predict a continuous target variable using one independent variable. It assumes that there is a linear relationship between the input and output.

For example, we can predict salary based on years of experience. The basic equation is:



EXAMPLE Salary can be modeled as a function of years of experience when the relationship is approximately linear.

04 List and explain the Assumptions of Linear Regression

Linearity: There should be a roughly linear relationship between the independent variable(s) and the dependent variable.

Independence: Observations should be independent of each other. One observation should not strongly depend on another observation.

Homoscedasticity: The variance of the errors should remain approximately constant across different predicted values.

01 Linearity Inputs and target follow a roughly straight-line relationship.	02 Independence Observations do not strongly depend on each other.
03 Homoscedasticity Error variance stays approximately constant.	04 Normality of Residuals Residuals are approximately normally distributed.
05 No Strong Multicollinearity Predictors are not highly correlated.	06 No Significant Outliers Extreme points do not distort the model.

Six conditions that support a reliable linear-regression model

Normality of Residuals: The residuals, which are the differences between actual and predicted values, should be approximately normally distributed, especially when we want reliable statistical inference.

No Strong Multicollinearity: In multiple linear regression, independent variables should not be highly correlated with each other.

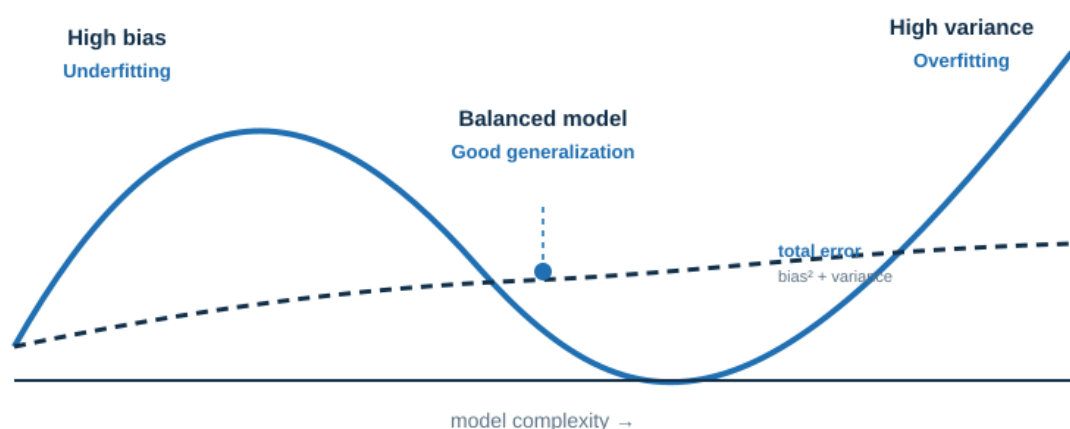
No Significant Outliers: Extreme observations should not strongly affect or distort the regression line and model coefficients.

05 What is Bias–Variance Trade-Off?

Bias is the error caused when a model is too simple and cannot capture the actual patterns in the data.

Variance is the error caused when a model is too sensitive to the training data and learns noise along with the actual patterns.

The Bias–Variance Trade-Off means finding the right balance between bias and variance. A good model should have low bias and low variance so that it can learn important patterns without overfitting.



IMPORTANT Increasing model complexity can reduce bias but increase variance. The best point is the one that generalizes well to unseen data.

06 Explain Overfitting and Underfitting with Examples

Overfitting occurs when a model learns the training data too well, including noise and unnecessary patterns. It usually performs very well on training data but performs poorly on new or unseen data. For example, a very complex Decision Tree may memorize the training examples instead of learning general patterns.

Underfitting occurs when a model is too simple to learn the important patterns in the data. It performs poorly on both training and testing data. For example, using a simple straight-line model for data that clearly follows a complex curved relationship.



KEY IDEA A useful model is complex enough to learn meaningful structure, but not so complex that it memorizes noise.

KEY TAKEAWAYS

Supervised Learning — At a Glance

The six concepts in this section connect the learning setup, the prediction task, the linear model, its assumptions, and the role of model complexity.

✓	Supervised learning uses labeled data.
✓	Regression predicts continuous numerical values.
✓	Classification predicts categories.
✓	Linear Regression models relationships between variables.
✓	Regression has important assumptions.
✓	Bias and variance must be balanced.
✓	Underfitting means the model is too simple.
✓	Overfitting means the model is too complex.
FINAL NOTE Good supervised-learning practice combines clear labels, an appropriate algorithm, validated assumptions, and a model complexity that supports generalization.	